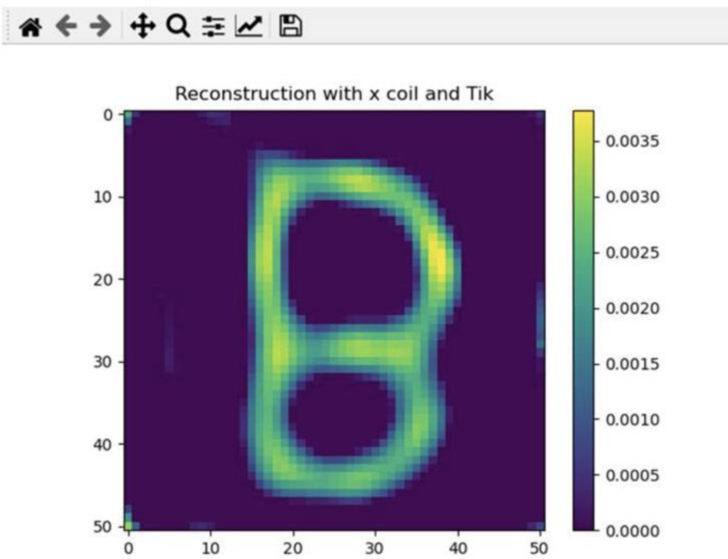
Dear Prof. Elfimova,

I am Zhong Hangyu, a doctoral student under the supervision of Professor Zhong Jing. Below is an introduction of our in-house MPI dataset and instruction on how to construct a training dataset for supervised neural networks.

You can download our in-house MPI dataset using this link:
https://drive.google.com/drive/folders/1gU-YJzdcs-BiJayFE8_V3xA-Er40EZVW?usp=sharing

After downloading and unzipping the file "Project.zip", you will find a folder named "DATA", a .docx file, a .pdf file, and along with some .py files. Here is what each file means:

- **DATA folder**: It contains eight .h5 files. These files store measurement data and system matrix data. For the details of the data structure, please read "H5_file_parameter_definition.docx".
- **H5_file_parameter_definition.docx**: This file defines the data structure of the .h5 files. It is modified from the MDF format (Magnetic Particle Imaging Data Format) proposed by Knopp et al. (see: https://arxiv.org/abs/1602.06072).
- **Three .py files** (MAIN.py, ReconstructionMethod.py, Nargout.py): You can try running "MAIN.py" It will read the system matrix data and measurement data from the DATA folder based on the filename you specify, and then reconstruct an image using Tikhonov regularization method. If it runs successfully, you should see an image like this:



You can also try changing the line: " filenameMeas = r'.\DATA\MeasurementData_B.h5' " to reconstruct images based on other data files.

- **Reference_Mixing-Space Magnetic Particle Imaging.pdf**: This is the reference paper for our in-house dataset. It includes a detailed description of the MPI data acquisition setup.

If you want to use the system matrix collected by our MPI device to build a dataset for training a supervised neural network, you can follow steps:

1. **Extract the system matrix**: In "MAIN.py", find the variable "SM", which represents the system matrix. Its size is 2550×2601. 2550 is the number of measurement samples. $2601 = 51 \times 51$ is the number of pixels in the reconstructed image.
2. **Generate the dataset**: Use computer simulation to create N different 51×51 images. Reshape each image into a 2601×1 vector. Multiply each vector by the system matrix SM to get a 2550×1 measurement vector. This gives you N pairs of "measurement vector - ground truth image", which form your training dataset.
3. **Train the network**: During training, feed each 2550×1 measurement vector into the neural network. The network outputs a predicted image. Compare this prediction with the true image and compute the error. Use backpropagation to update the network's parameters until the network converges.
4. **Run inference**: To reconstruct real-world data (e.g., measurement data from the "DATA" folder), input the real-world 2550×1 measurement vector into the trained network. The well-trained network's output will be the high-quality reconstructed image.

Note that this workflow is for the supervised network that directly maps from measurements to images. Currently, various types of neural networks are proposed, and corresponding modifications are needed for specific tasks.

We hope the details above are helpful.

Sincerely,
Hangyu Zhong